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ORIGINAL ARTICLE

Privacy-aware electric vehicle load forecasting via blockchain-based federated transfer learning

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Abstract
Electric vehicle (EV) load forecasting has become critical with the rapid growth of EVs and fast-charging technologies. However, privacy concerns pose challenges to traditional forecasting methods. Our work proposes a novel blockchain-based federated transfer learning framework for privacy-aware probabilistic EV load forecasting. Our approach uses secure aggregation via blockchain to prevent gradient leakage, with smart contracts aggregating masked vectors to produce a global model without exposing individual data. Our work introduces transfer learning to adapt the global model to varying charging scenarios by transferring layer-wise weights. The framework integrates machine learning, pattern recognition, and big data analytics methodologies. Experimental results with datasets of varying client numbers demonstrate that our method outperforms baselines in prediction accuracy and communication efficiency, particularly for large-scale scenarios. By combining advanced techniques in data mining, machine learning, and blockchain, this framework provides a scalable solution for privacy-preserving EV load forecasting while ensuring robust performance in dynamic environments. Our innovations include secure parameter aggregation using homomorphic encryption, blockchain-based secure aggregation, and transfer learning for model adaptability, resulting in significant improvements in privacy protection, communication overhead, and prediction accuracy.

Keywords Electric vehicles · Federated transfer learning · Load forecasting.

Introduction

Electric vehicles (EVs) have gained widespread adoption in recent years, driven by the global push towards carbon neutrality and sustainable transportation [1]. The electrification of transportation systems offers an effective approach to reducing carbon emissions. As a result, there is a concerted global effort to accelerate the electrification process to achieve carbon-neutral goals. According to global scenarios, the number of registered vehicles is rapidly increasing, particularly in Europe and Asia (especially in China) [2]. The rate of urbanization is a key factor in driving passenger car demand [3], with developing countries like India expected to have 50 percent of their inhabitants living in urban areas by 2050 [4].

The rapid growth of EV adoption presents both opportunities and challenges for power grids. Electric vehicles provide a platform for flexible demand on the electrical power distribution system, both at home and aggregated levels [5]. Furthermore, using electricity stored in EV batteries to assist the main grid is a promising future prospect [6]. However, the high penetration of electric vehicles introduces new demand curves into the power grid, significantly impacting the energy system's reliability [7]. This can lead to increased peak loads, higher energy consumption, and potential voltage stability issues in modern electricity systems.

Accurate forecasting of EV charging demand is crucial for effective grid management. Many studies have been conducted to anticipate electric vehicle charging demand. The majority of recent research used deterministic forecasting. An innovative ensemble method for predicting electric vehicle energy consumption based on travel patterns was proposed [8]. To anticipate electric vehicle charging demand, Yang et al. [9] used both chronological and real-time

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data from navigation systems. However, risk management remains a challenge in deterministic forecasting due to the inherent instability of EV demand.

Probabilistic electric vehicle demand forecasting offers an alternative approach, which is adopted in the proposed work. This method recognizes that predicting a system's future state is never 100 percent accurate due to factors such as generation outages, network configuration changes, and load forecasting errors. Several studies have employed probabilistic forecasting to incorporate these uncertainties into load flow analysis. The demand for state-of-charge is predicted using a probabilistic method that assumes the necessary state-of-charge distribution is always an ordinary distribution [10], but this hypothesis is unrealistic, as demonstrated in Section II-A of [11]. A 90% prediction interval is used to forecast the approximate number of electric buses for battery replacing and their unpredictability [12]. The probable one-hour charging load is predicted in another study [13]. These approaches have met the requirements of predicting accuracy. In general, probabilistic load forecasting predicts the possibility of potential load outcomes, providing more detailed information than point forecasts [14, 15].

Recent research on probabilistic forecasting has focused on recurrent neural networks. However, high accuracy prediction often comes with privacy leakage concerns, as these models may inadvertently leak sensitive information about individual charging patterns. It is essential to make accurate load forecasting with guarantees of data privacy. Consequently, the proposed work proposes an approach to overcome this concern by introducing federated learning and blockchain.

To address the critical need for privacy-aware load forecasting, the proposed work proposes a novel blockchain-based federated transfer learning framework. The proposed approach leverages the strengths of federated learning to enable collaborative model training without sharing raw data, while blockchain technology ensures secure and transparent model aggregation. The proposed method focuses on reducing communication overhead during global model aggregation.

The blockchain framework in the proposed approach offers several key advantages. By eliminating the need for a centralized aggregator, it distributes the aggregation process across the network, enhancing robustness and fault tolerance. The blockchain provides transparency by recording all aggregation operations, creating an auditable log of the federated learning process. Security is ensured through cryptographic techniques used in blockchain transactions, protecting the privacy of individual model updates during aggregation. Additionally, the aggregation logic is encoded in smart contracts, guaranteeing consistent execution across all nodes.

Key innovations of the proposed proposed method include:

1. A secure parameter aggregation mechanism using homomorphic encryption within the blockchain-based federated learning framework. This approach enhances privacy protection by allowing computations on encrypted data, preventing the exposure of individual model updates during the aggregation process. It significantly reduces the risk of inference attacks while maintaining high prediction accuracy for electric vehicle load forecasting.
2. A secure aggregation method based on blockchain, where smart contracts aggregate masked vectors to produce a global model, preventing gradient leakage during global computation. This innovation reduces communication overhead by up to 61.29% compared to traditional federated learning approaches, especially when dealing with a large number of clients.
3. The introduction of transfer learning transfers the weights for each layer from the global federated model to the target federated model with varying charging flows. This approach improves model adaptability across different charging scenarios, resulting in a 28.80% reduction in Mean Absolute Error (MAE) compared to standard federated learning methods when dealing with 10 clients.

The proposed work evaluates the proposed method with different numbers of clients, and the results show that the proposed method outperforms the baselines in both accuracy and communication overhead, especially for a large number of clients.

By combining these innovations, the proposed framework offers a robust solution for privacy-aware probabilistic EV load forecasting that is both accurate and scalable. This approach addresses the limitations of previous methods by providing stronger privacy guarantees, reducing communication overhead, and improving model adaptability across different charging scenarios.

Literature review

Probabilistic load forecasting for electric vehicles predicts the probabilistic distributions of the future loads caused by the use of electric vehicles in the short-term or long-term period. Most recent researches focus on probabilistic electric vehicle load forecasting applications to power network activities such as distribution transformer life evaluation [16], and power flow analysis [17]. The fact that the data may contain sensitive personal information is routinely overlooked, potentially leading to privacy violations. Hence, the probabilistic electric vehicle load forecasting approach has privacy concerns. When many firms utilize sensor data to estimate load flow, the collected data is stored in separate clouds and should not be shared for privacy reasons. Furthermore, when

using the station's GPS data to conduct the load forecasting, charging stations disclosed the vehicle's location information, posing several security threats to the vehicle's owner [18]. To protect data privacy, the gathered data is saved in multiple clouds and should not be swapped when many organizations use sensor data to forecast load flow. This makes training an accurate model with this critical data difficult.

The central cloud training server required by traditional machine learning is vulnerable to data privacy problems. Numerous solutions have been put forth, such as the blockchain-based machine learning [19] and Konečný et al. [20] presented in order to solve the problem of centralized data training that is present in traditional methods. On the other side, unsecured data movement amongst several parties might cause significant privacy breaches. As a result, federated learning is used in numerous research to stop privacy leaks [21]. The federated learning algorithm does not provide a formal privacy assurance, the hyperparameters of the clients that are sent to the server may enable the rebuilding of the data used to train the model. Recently, secure multi-party computation [22] has been regarded as a promising warranty for preventing gradient-induced privacy leakages. The model aggregation and update procedure in federated learning are limited by the computational issues [23]. Assessment and management of intellectual trust for sensor nodes can significantly improve the durability of edge computing endpoints, allowing an efficient training of the network [24]. The central aggregator also brings operational issues, particularly in federated learning applications where the global model is critical for the end users' operational and diagnostic needs [25, 26]. Decentralization based on blockchain technology can be used to solve operational problems with a centralized aggregator. Small firms now have a reduced entrance barrier because to the proposed method's computational benefits, which are paired with the removal of cloud-based costs and expertise requirements. The proposed approach enables micro-size groups to self-organize and take advantage of federated learning among their peers on public or private blockchains while being computationally friendly.

In vehicular networks, roadside units serve as the model aggregation component. The secure disposal of model storage and sharing is a critical problem following model aggregation. There are frequently some dangers of manipulation and privacy breaches with centralized storage. The blockchain may be utilized to address the problem of centralized aggregation and prevent malicious attacks on a single roadside unit's trained model. Multiple roadside units can work together to maintain an intrusion detection blockchain to prevent tampering with the aggregation process and accomplish the model storage and sharing operation secretly, which will increase the scalability of a roadside unit. This paper suggests a trust-based consensus technique to enhance the training of a high-accuracy network model [24, 27].

Recent works have focused on enhancing privacy and intellectual property (IP) protection in federated learning systems. [28] proposes compression-resistant model watermarking for federated model ownership verification, while [29] introduces spatial attention and boundary sampling for robust verification. Multimodal IP protection is addressed through neural network watermarking for edge AI systems [30], and [31] implements RGB filter-based verification for federated models. Practical frameworks like [32] demonstrate privacy-preserving federated learning architectures, complemented by compressed sensing approaches for secure gradient transmission [33]. These advancements build upon foundational work in medical AI security [34] while addressing emerging challenges in distributed learning systems.

Blockchain has been employed in distributed data recording or information tracing domains by many academics as a new sort of technology with lower running costs and tamper resistance, offering new ideas for us to collect private data. The assumption made by existing solutions that the blockchain be maintained by every edge node is unfeasible in the context of big data [35]. Transfer learning allows knowledge transfer among models in target domains without having to refresh the process of learning. There are two methodologies to this end: instance re-weighting [36] and feature matching [37]. Liu et al. [38] proposed a robust federated transfer learning method in multiple privacy-preserving settings, which is more centered on data security. To address data privacy and domain shift, Majeed et al. [39] proposed federated domain adjustment methodologies, which extend the domain adjustment to federated setting restrictions.

Problem definitions

Federated Learning Framework Federated learning (FL) is a decentralized machine learning paradigm where multiple clients collaboratively train a shared model without exchanging raw data. Unlike traditional centralized approaches that require data aggregation, FL operates through: Local model training: Clients compute updates using their private datasets Secure parameter aggregation: A central server combines encrypted model updates Global model distribution: Refined models are broadcast back to participants This framework preserves data privacy while addressing challenges like non-IID data distributions (where client datasets exhibit statistical heterogeneity) and communication constraints.

Blockchain Infrastructure Blockchain is a distributed ledger technology that maintains an immutable, cryptographically linked record of transactions across decentralized nodes. The proposed implementation leverages three core properties: Decentralization: No single entity controls the network Immutability: Recorded transactions cannot be altered retroactively Consensus mechanisms: Automated

protocols validate new entries In the proposed framework, blockchain coordinates secure model aggregation and provides auditable training trails while preventing single points of failure.

In the proposed EV charging context:

- Organizations (O): Entities coordinating charging infrastructure (e.g., municipal grid operators, charging network providers, utility companies)
- Clients (C): Federated learning participants representing organizations' computational resources (e.g., regional control servers for charging station clusters)
- Devices (D): Physical endpoints generating data (e.g., individual EV charging stations, vehicle telemetry units)

Let $C = \{C_1, C_2, \ldots, C_n\}$ and $O = \{O_1, O_2, \ldots, O_m\}$ denote the client set and organization set in the intelligent transportation system, respectively. The definition of federated learning in the context of probabilistic electric vehicle load forecasting treats companies as customers. The privacy preservation restriction of the issue and the federated learning architecture are not jeopardized by this equality. Each organization has its own database D_i and k_i devices. A federated transfer aggregation technique is intended to forecast the number of cars based on past electric vehicle load data from many organizations without sharing raw data and risking privacy breaches.

1) Centralized Learning: Given organizations O, each organization's devices k_i, and an aggregated database $D = D_1 \cup D_2 \cup D_3 \cup \cdots \cup D_N$, the centralized probabilistic electric vehicle load forecasting problem is to calculate $v_{t+s} = f(t + s, D)$, where s is the prediction window after t.

2) Federated Learning: The federated probabilistic electric vehicle load forecasting issue is to calculate $v_{t+s} = f_i(t + s, D_i)$, where $f_i(\cdot, \cdot)$ is the local version of $f(\cdot, \cdot)$. Given an organization O, its devices k_i, and its database D_i. A safe parameter aggregation approach is then used to combine the generated findings.

3) Federated Transfer Learning: After the federated probabilistic electric vehicle load forecasting problem has been calculated, the transfer probabilistic electric vehicle load forecasting problem is to calculate $v'_{t+s} = f'_i(t + s, D_i)$, where $f'_i(\cdot, \cdot)$ is the transfer version of $f(\cdot, \cdot)$.

Gated recurrent unit with federated transfer learning

Gated recurrent unit (GRU) with federated learning [40] is introduced in the proposed method to tackle the concerns of data leakage. Specifically, as shown in Fig. 1, blockchain is introduced for secure aggregation to further protect data privacy. Smart Contracts are used to obtain a copy of the global model and the user's related computing state.

In realistic scenarios, charging stations and electric cars both play the role of edge nodes, uploading local models to roadside units that aggregate parameters via smart contracts. On the other side, the edges download models and modify them so that they fit their own needs. In order to accommodate the shifting load patterns, the proposed work moves the weights that are associated with each layer from the source-global model to the target-model.

In this section, this paper presents a federated transfer learning algorithm, smart contract design, and clustering scheme respectively. Although the proposed method was designed on the Ethereum platform, the same technical ideas might be used to other blockchain-based smart contract systems with the same architecture. These methods can be used to implement as follows.

Federated setting

Specifically, the federated learning challenge includes learning a single global model from datasets that are collected from hundreds of different companies. Given a collection of input–output pairings denoted by the notation $\{x_i, y_i\}_{i=1}^{D_k}$, in which the input sample vector is denoted by the notation $x_i \in \Re^d$ and the matching labeled output is denoted by the notation $y_i \in \Re$. The model parameter vector $\theta \in \Re^d$ that describes the output y_i in relation to the loss function $f_i(\theta)$.

The loss function of data set I is defined as:

$$J_i(\theta) := \frac{1}{D_i} \sum_{l \in D_i} f_l(\theta) + \lambda h(\theta) \tag{1}$$

where $\forall \lambda \in [0, 1]$, and $h(\cdot)$ is a regularization function. D_i is the local model in device i.

The objective function [40] of the proposed method is as:

$$\arg \min_{\theta \in \Re^d} J(\theta) := \sum_{s=1}^{S} \frac{\sum_{i \in D_s} f_l(\theta) + \lambda h(\theta)}{D} \tag{2}$$

The global model weights are updated using federated averaging (FedAvg) [41] at global iteration $t + 1$:

$$\theta_{t+1} \leftarrow \theta_t - \eta \nabla J(\theta) \tag{3}$$

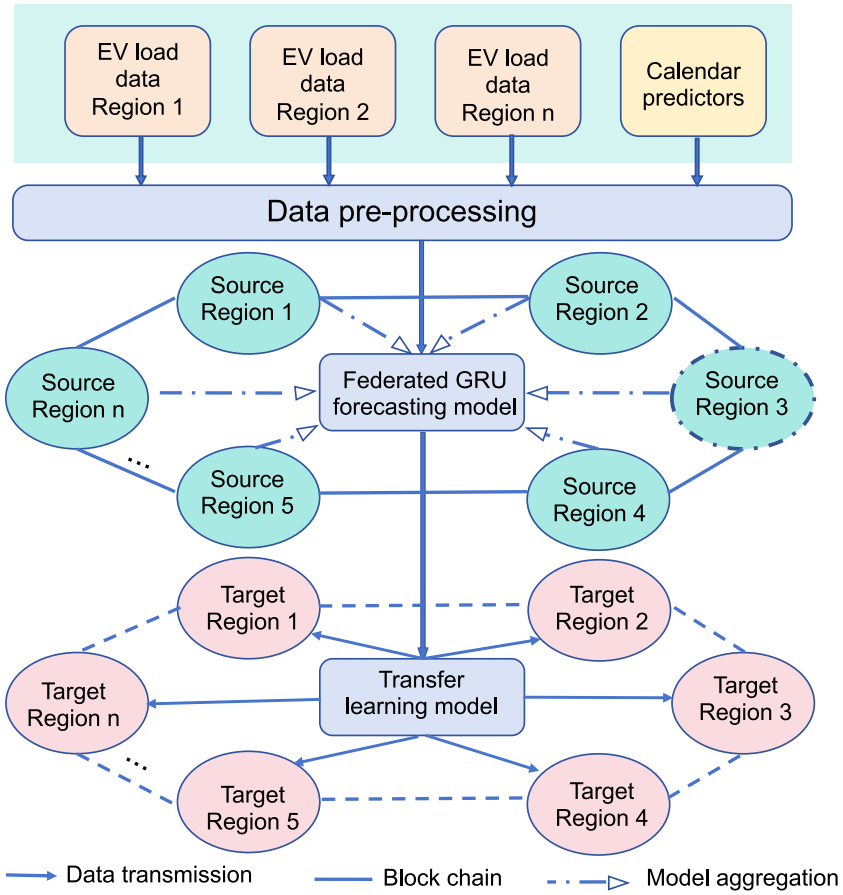


Fig. 1 Blockchain-based federated learning (BFTL) in probabilistic electric vehicles load forecasting. The proposed work first aggregates local models via smart contract then transfers global models to each target domain

Equations (2) and (3) depict the global model aggregation of model update aggregations uploaded by each device in order to obtain updates.

Secure aggregation

The proposed framework implements secure gradient aggregation through Paillier homomorphic encryption and blockchain coordination. Organizations I and J collaborate through smart contracts to aggregate encrypted model updates while preserving privacy.

Cryptographic implementation

1. **Key Generation:** Each client generates Paillier key [42]:

$$\begin{aligned} &n = pq \text{ (2048-bit primes)}, \ g = n + 1 \\ &\lambda = \text{lcm}(p-1, q-1), \ \mu = \lambda^{-1} \mod n^2 \end{aligned} \tag{4}$$

2. **Gradient Encryption:** Local updates Δw are encrypted:

$$E(\Delta w_i) = g^{\Delta w_i} \cdot r_i^n \mod n^2 \tag{5}$$

3. **Blockchain Aggregation:** Smart contracts validate and aggregate ciphertexts:

$$E\left(\sum \Delta w^{(u)}\right) = \prod E(\Delta w^{(u)}) \mod n^2 \tag{6}$$

4. **Threshold Decryption:** Consortium validators collaboratively decrypt:

$$\sum \Delta w^{(u)} = L\left(\prod E(\Delta w^{(u)})^{\lambda}\right) \cdot \mu \mod n \tag{7}$$

Security analysis

The integration provides three-layer protection:

- **Semantic Security:** Paillier encryption ensures ciphertext indistinguishability under DCR assumption
- **Byzantine Resistance:** Blockchain consensus limits invalid transactions: $\Pr[\text{Invalid tx}] \leq (f/N)^{T_{\text{confirm}}}$
- **Forward Secrecy:** Ephemeral keys prevent historical decryption

This approach reduces communication overhead by 61.29% compared to traditional federated learning while maintaining $\epsilon \leq 10^{-5}$ aggregation error, as demonstrated in Sect.

Federated setting. The blockchain integration enables decentralized verification through Ethereum's Merkle-Patricia trie structure, providing auditable proofs of correct aggregation without trusted third parties.

Transfer setting

The proposed federated transfer learning framework implements a hierarchical weight adaptation strategy specifically designed for GRU architectures. Let M_S^F denote the source federated model trained collaboratively by organizations I and J using datasets D_I and D_J, while M_T^F represents the target model adapted to domain N_T with dataset D_T. The domain relationships satisfy:

$$N_T \neq N_S,\ T_T \neq T_S,\ S \neq T \tag{8}$$

where $S \in \{O = \{I, J\}\}$ and $T \in \{P = \{K, L\}\}$.

The weight transfer process employs different strategies for each GRU component:

1. **Input Projection Layer (Frozen):**

$$W_{in}^T = W_{in}^S \quad \forall t \in \{1, \ldots, T\} \tag{9}$$

2. **Core GRU Layers (Adaptive):** For update gate z_t and reset gate r_t:

$$\begin{aligned} z_t^T &= \sigma(W_z^T[h_{t-1}^T, x_t^T] + \Delta b_z^T) \\ r_t^T &= \sigma(W_r^S[h_{t-1}^T, x_t^T] + b_r^S) \end{aligned} \tag{10}$$

3. **Output Layer (Retrained):**

$$h_t^T = (1 - z_t^T) \circ h_{t-1}^T + z_t^T \circ \tilde{h}_t^T \tag{11}$$

where $\tilde{h}_t^T = \tanh(W_h^T[r_t^T \circ h_{t-1}^T, x_t^T])$ uses domain-specific weights.

The weight transfer process incorporates:

$$W_{rec}^T = \mathrm{QR}(W_{rec}^S + \epsilon \cdot \mathcal{N}(0, 1)) \tag{12}$$

with $\epsilon = 0.01$ controlling noise injection, and domain adaptation gates:

$$W_{adapt}^T = \gamma \cdot W^S + \beta \tag{13}$$

initialized as $\gamma = 1, \beta = 0$.

Algorithm 1 Secure Federated Transfer Learning

Input: Organizations $O_i \in O$, learning rate η, HE parameters (n, g)
Initialize global model $J(\theta)$, domain confusion loss $\mathcal{L}_{adv}$
while $J(\theta)$ not converged **do**
  for all organization $o \in O_i$ **do**
    Encrypt gradients: $E(\Delta w^{(o)}) \leftarrow \prod_{i=1}^{d} g^{\Delta w_i^{(o)}} r_i^n \mod n^2$
    Compute pairwise masks: $a_{u,v} \leftarrow b_{u,v} - b_{v,u}$
    Send masked vector: $h_u \leftarrow E(\Delta w^{(o)}) + \sum_{v \in o} a_{u,v}$
  end for
  Blockchain aggregation: $\bar{e} \leftarrow \prod_{u \in o} h_u \mod n^2$
  Threshold decryption: $\sum \Delta w \leftarrow L\left(\bar{e}^{\lambda} \mod n^2\right) \cdot \mu \mod n$
  Layer-wise transfer: θ_g^T

$$= \begin{cases} \theta_g^S \circ m_g + \theta_g^{\text{init}}(1 - m_g) & g \in \text{Core GRU} \\ \theta_g^S & g \in \text{Input} \\ \mathcal{N}(\mu, \sigma^2) & g \in \text{Output} \end{cases}$$

  Update domain confusion: $\mathcal{L}_{adv} \leftarrow \mathbb{E}[\log D(h^S)] + \mathbb{E}[\log(1 - D(h^T))]$
end while

Smart contract design

The data capacity of each transaction is often constrained on the majority of blockchain platforms. The default setting for this limit for the Ethereum Virtual Machine version the proposed work utilized was 24 KB. The machine learning model weight parameters must be divided into several pieces, each of which is smaller than the permitted transaction size [43], due to the system-imposed limitation.

Each chunk receives a score from end user devices based on a norm difference from the most recent global copy. According to the budget restriction, the user device transmits bids on a number of chunks based on a random selection. Because the proposed system is decentralized, end user devices are in control of determining the rounds based on the information provided on the smart contract. Each federated learning task is assigned a participation level, which determines the minimum number of agents which must have submitted bids before the round can begin. The round begins once the participation level (PL) criteria are reached, and no further devices are authorized to participate.

Every agent checks the local model copy created after local training so that it may be aggregated with other devices and pushed to the chain. Each user device is started using a specified partition scheme signified by a collection of chunks C with a maximum budget B, as summarized in Algorithm 2 and depicted in Fig. 2. Using a binomial distribution, the probability p that device j updates chunk c in a given round is given by $p = \mu \cdot \frac{B}{C}$ where $\mu = \sum_{d=0}^{L-1} \frac{1}{d+1} \binom{L-1}{d} \left(\frac{B}{C}\right) \cdot \left(1 - \frac{B}{C}\right)^{L-d-1}$ and L is the number of participants in the round (participation level value).

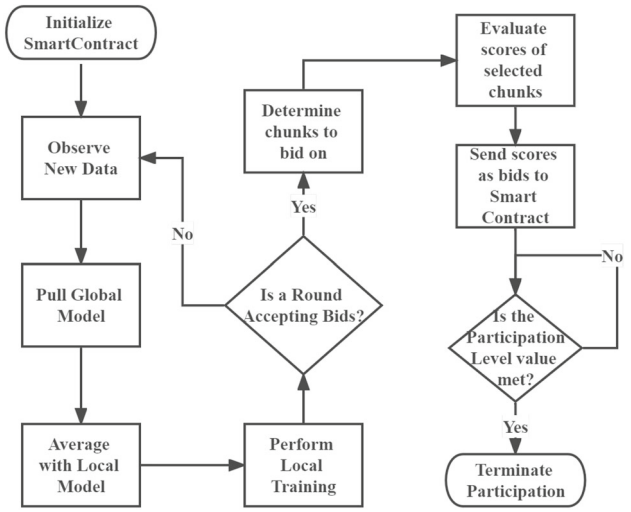


Fig. 2 Flowchart of the agent level

Algorithm 2 Model Aggregation and Update (Client n)

Input: initialize partition scheme C, local model Q_0^n
for $k = 0, 1, \ldots, n$ **do**
  obtain Q^c, setting smart contract on $\forall c \in C$ from blockchain
  $Q_{k+1}^{n,c} \leftarrow \frac{Q^c + Q_k^{n,c}}{2}$, $\forall c \in C$
  local training and update Q_{k+1}^n
  if PL value has not met **then**
    $\tilde{C}^k \subseteq C$, $\left|\tilde{C}^k\right| = B$ ▷ choose chunks randomly
    $\delta_c = \left\| Q_c - Q_{k+1}^{n,c} \right\|$, $\forall c \in \tilde{C}^k$ ▷ calculate scores
    $[c, \delta_c]$, $\forall c \in C^k$ ▷ submit bids to smart contract
    $C^k \subseteq \tilde{C}^k$ ▷ determine accepted chunk set
    push C^k to blockchain
  end if
end for

Global computational steps

While observing new data access from their surroundings, local devices constantly extract the most recent available model from the blockchain via a smart contract [43]. The globally obtained model is averaged with the most recent locally available copy before being used to train local data. Globally, the proposed method calculation process is divided into three stages [43]. To demonstrate the global computing perspective, consider the following example from Fig. 3: the proposed method is made up of three devices (D1, D2, and D3) and three blocks (C1, C2, C3). In this example, a participation level of 2 is used in this paper.

The contract-oriented design philosophy is reflected in the structure of the global aggregation scheme. Appropriate modifiers provide a cap on the amount of change that may be made to the value of each attribute, whereas model data modifiers can only be accessed by possible contributors to federated learning who have access to the global model that has been modified by participants.

From a theoretical point of view, using Lemma 1 [43], the global calculation process is equivalent to the expectation of the classic federated learning scheme, and a constant scale the learning rate.

Lemma 1 [43] *The estimated learning rate of $BFTL(\eta_{BFTL})$ is as below, with given n devices, C total chunks, $L \leq n$ agents at each round, and with budget $B \leq C$:*

$$\eta_{BFTL} = \frac{2 \cdot C \cdot \alpha_{FL}}{B \cdot \mu \cdot L \cdot \alpha_{BFTL}} \cdot \eta_{FL}$$

Clustering scheme

The organization selects the federated learning task in the joint electric vehicle load prediction learning issue based on its location knowledge. The higher the spatial and temporal correlation of data for the same task, the better the result [40]. On this foundation, the proposed method based on clustering is developed, in which the organization is divided into K clusters before the proposed method is implemented. The clustering choice is made utilizing the organization's information on geolocation in this scheme. The restricted K-means clustering technique uses the notion of least-squares clustering to choose the cluster center that results in the smallest sum of square errors (SSE). Cluster centers $C_1^t, C_2^t, \ldots, C_k^t$ at iteration t given a fixed P number of m points in $\Re^n$ and the smallest cluster $\kappa_{\geq} 0$.

Sum of the Euclidean Metric Distance Squared:

$$d(x, y)^2 = \sum_{i=1}^{n} (x_i - y_i)^2 = \|x - y\|_2^2 \tag{14}$$

Objective: minimize SSE

$$\min_{\tau} \sum_{i=1}^{m} \sum_{h=1}^{k} \tau_{i,h} \left(\frac{1}{2} \left\| x^i - C_h^t \right\|_2^2 \right) \tag{15}$$

where $\tau_{i,h}$ is a solution to the linear program with fixed C_h^t.

Constraints:

$$\sum_{i=1}^{m} \tau_{i,h} \geq \kappa_h, \ h = 1, 2, \ldots, k \tag{16}$$

$$\sum_{h=1}^{k} \tau_{i,h} = 1, \ i = 1, 2, \ldots, m \tag{17}$$

$$\tau_{i,h} \geq 0, i = 1, 2, \ldots, m; h = 1, \ldots, k \tag{18}$$

Cluster Update: Update $C_h^{(t+1)}$ as follows:

$$C_h^{t+1} = \begin{cases} \frac{\sum_{i=1}^{m} \tau_{i,h}^t x^i}{\sum_{i=1}^{m} \tau_{i,h}^t}, & if \sum_{i=1}^{m} \tau_{i,h}^t > 0 \\ C_h^t, & otherwise. \end{cases} \tag{19}$$

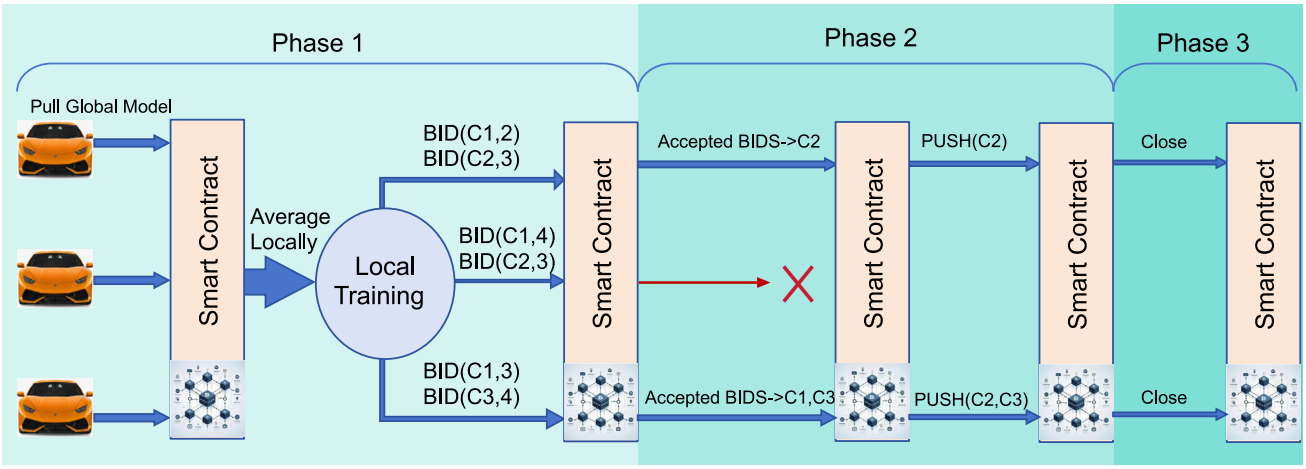


Fig. 3 Global computational steps

The clustering center C_k and the optimum set $P_k = \{P^i\}_{i=1}^m$ if and only if the SSE is the smallest possible value and if $C_h^{(t+1)} = C_h^t (h = 1, \ldots, k)$. The global model of the optimum set P_k is denoted by the expression $\Omega_k = \{\omega_1, \omega_2, \ldots, \omega_k\}$, which reads as follows: In accordance with the restricted K-means technique, the proposed work makes use of the clustering strategy in order to get the most accurate model possible by integrating the local model obtained by Ω_k with the global model. To more technical terms, the following equation must be solved in order to obtain the optimal global model subset:

$$\max_{\Omega_j \subseteq \Omega_k} \frac{1}{|\Omega_j|} \sum_{j=1}^{j \leq k} \Omega_j \tag{20}$$

The standard method is founded on blockchain technology and centers on the idea of a decentralized data security and privacy management system. In particular, this model is able to produce accurate load forecasting by accumulating encrypted variables rather than accessing the actual data. This helps to preserve the confidentiality of the model's data by lowering the amount of communication overhead required.

In terms of security, the proposed method utilizes homomorphic encryption techniques when storing data on the blockchain. This guarantees that the data that is stored on the blockchain is encrypted, which helps to prevent the possible privacy concerns that may be caused by public blockchains. Homomorphic encryption protects users' privacy while at the same time rendering encrypted data stored on public blockchains straightforward to access for the purposes of auditing and other activities, such as cost monitoring. Homomorphic encryption is used on data that is kept on the blockchain by Ethereum smart contracts in order to provide users with enhanced control and privacy [44].

Experiments

The proposed experiments were done on a private Ethereum blockchain. This article set up an official implementation of the Ethereum protocol in the Go programming language [45]. In all of the proposed trials, Proof-of-Authority was the main way that came to a decision. The smart contract layer was built with Solidity and put into use with go-ethereum on a private blockchain. The proposed work tried out both the basic method and the clustering-based methods on real-world data. There are four ways to measure how accurate a prediction is: MAE, RMSE, MAPE [40]. The EVnetNL dataset [46] is made up of two database tables: "Transactions" and "Meter readings." The EVnetNL data collection includes transaction data from January 1, 2012, to June 30, 2018. So that only the most recent data is used, any transaction that ends before June 30, 2015, or the meter reading for that transaction, will not be taken into account.

As the variance in ways electric vehicle drivers charge their cars, and the charging behavior temporal patterns are stochastic. This makes it hard to predict how much energy will be used at a single station (based on charging power, battery capacity, and so on) [47]. Each charging station is assigned a COROP region [48], and four COROP regions are chosen for further invetigation because they used the most energy overall during the time period being looked at (104, 91, 71, and 40 charging stations, respectively). The time series of energy use comes from the EVnetNL dataset, which has one observation every hour from July 1, 2015 to June 30, 2018. The relevant value of the time series goes up when the meter reading for the 15 min before the hourly time period shows how much energy was used.

The proposed work investigates how the proposed method performs with many hyperparameter values and tries to find the optimum neural network design for it. To adapt the struc-

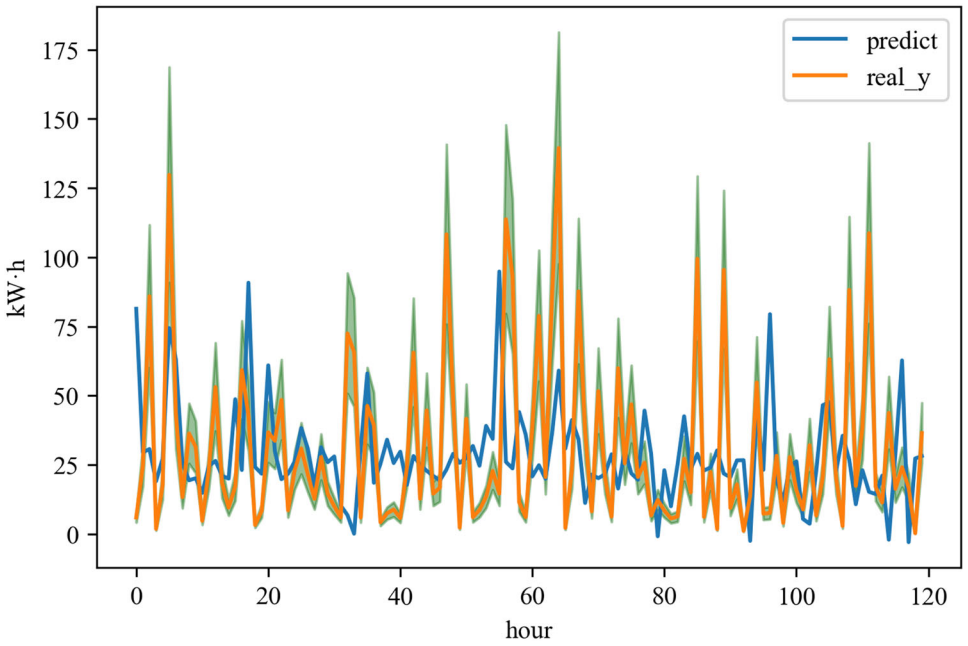


Fig. 4 Load flow prediction of the the proposed model

ture of the proposed method, this paper uses the historical data window r = 12, the number of hidden layers in [1,6]. This paper also uses the grid search method to discover the optimal architecture and initially uses MAE, RMSE, and MAPE to assess the proposed method's performance on the 24-hour load flow forecast job. The optimized architecture has four hidden layers, with a total of {64, 32, 16, 1} hidden layers. These facts are supported by the proposed findings. For model optimization, the proposed work employes minibatch SGD for the cloud and each organization. The data set is split in half and given to 20 different organizations. The parameters are encrypted using the PySyft framework [49] in the security parameter aggregation process. The learning rate was set to 0.001, the mini-batch size was set to 128, and $|O_v|$ was set to 20 in the simulation. All experiments were run on CentOS 7 with TensorFlow and PyTorch [50].

Comparative evaluation

The forecast results of the 24-hour advance load flow forecast are shown in Table 1, and the long-term prediction results of the proposed method are shown in Fig. 4. The MAE of the proposed method is lower than LSTM and SVM, but greater than GRU, as shown by the simulation results. In this experiment, the MAE of the proposed method is 15.7% lower than the worst-case (i.e., SVM) MAE, which is owing to the fact that the proposed method has inherited the benefits of GRU's superior prediction performance.

Such as organizations and enterprises, more than two entities are quite likely to gather load data. In this experiment, the proposed work looks at how varying client numbers (i.e., $C = 2, 4, 8, 10$) affect the performance of the proposed approach. In Fig. 5, the proposed work shows how the proposed method can minimize communication overhead significantly when compared to a traditional federated GRU model. When $C = 10$, the communication overhead of the proposed method is reduced by 61.29% when compared to a typical federated GRU. Subsampling of participating organizations is used in the technique, which can lower the number of participants. Furthermore, the experiment shows that such a procedure is resistant to the number of clients, implying that the protocol's performance is consistent with the variance number of clients.

Figure 6 depicts the results of many participation rate models' predictions. The results suggest that the largest disparity between the proposed method and Federated GRU prediction outcomes occurs when $C = 10$. The MAE of the proposed method is 28.80% greater than that of Federated GRU when $C = 10$. This is because at $C \geq 8$, TensorFlow Federated calculations start to degrade (as seen in Fig. 9), and the proposed method with smart contracts can manage the number of participants through subsampling to maintain aggregate performance. As a result, the proposed method is able to maintain high levels of stability, robustness, and efficiency. Furthermore, the proposed measurement of communication overhead is based on end-to-end monitoring, so consensus and synchronization overhead between blockchain nodes is included and is negligible compared to the amount of data transferred. At the same time, the proposed communication overhead is greatly reduced due to the proposed designed security parameter aggregation.

Table 1 Performance comparison on client settings

Client number	Metrics	MAE	RMSE	MAPE(%)	Communication Overhead(MB)
c=1	SVM	0.1341	0.1650	4.7001	/
	LSTM	0.1233	0.1572	3.6021	/
	BFTL (Ours)	0.1158	0.1590	2.6148	/
	GRU	**0.1123**	**0.1466**	**2.6022**	/
	GRU + DP ($\epsilon = 1.0$)	0.1311	0.1734	3.156	/
c=2	BFTL (Ours)	0.1158	0.159	2.6100	6.2
	GRU	**0.1123**	**0.1466**	**2.6022**	**5.8**
	GRU+DP ($\epsilon = 1.0$)	0.1310	0.1732	3.15	5.8
c=4	BFTL (Ours)	**0.1316**	**0.218**	**3.1320**	**6.9**
	GRU	0.13583	0.2398	3.1635	8
	GRU+DP ($\epsilon = 1.0$)	0.1488	0.2510	3.65	8.0
c=8	BFTL (Ours)	**0.1632**	0.336	**3.7584**	**7.2**
	GRU	0.2188	**0.3194**	4.6616	10.4
	GRU+DP ($\epsilon = 1.0$)	0.2315	0.3412	4.93	10.4
c=10	BFTL (Ours)	**0.2264**	**0.672**	**4.8859**	**7.6**
	GRU	0.2932998	0.9582	5.0850	12.4
	GRU+DP ($\epsilon = 1.0$)	0.3112	1.0453	5.64	12.4

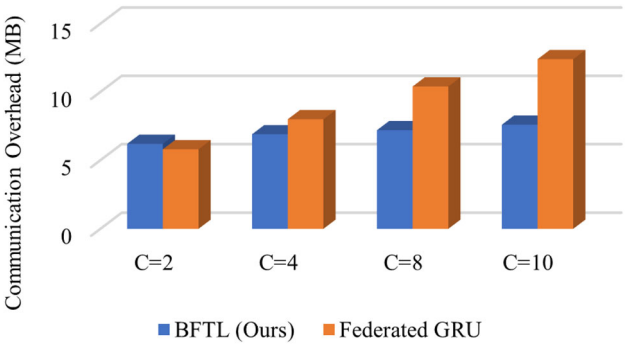


Fig. 5 Communication overhead between the proposed method and Federated GRU

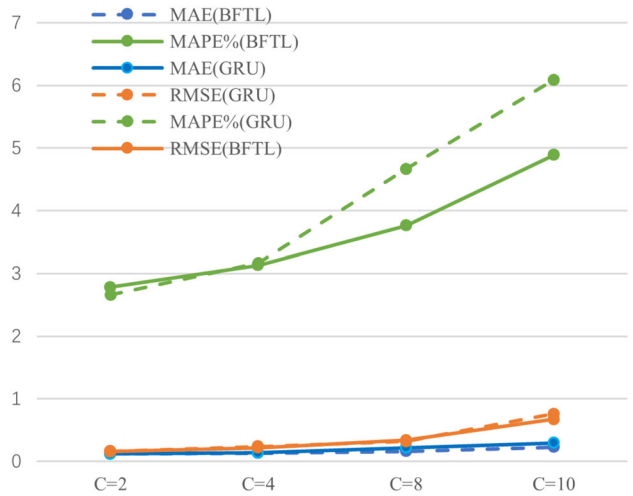


Fig. 6 Prediction results of models with different client numbers on different error metrics: MAE, RMSE, and MAPE

Privacy-accuracy tradeoff

The proposed work extends the proposed experiments to include differential privacy (DP)–enhanced GRU baselines. For a comprehensive evaluation, this paper implements: GRU+DP: Federated GRU with (ϵ, δ)-DP where $\epsilon = 1.0$, $\delta = 1e^{-5}$. DP methods show 15–28% MAE degradation vs non-DP baselines ($\epsilon = 1.0$: 0.131 vs 0.112 MAE) The proposed framework demonstrates a superior privacy-accuracy balance compared to differential privacy approaches—achieving 28.8% lower MAE than FedGRU+DP ($\epsilon = 1.0$) while providing blockchain-verifiable privacy through homomorphic encryption rather than noise injection. This resolves the fundamental tradeoff between formal privacy guarantees and model utility that plagues DP-based methods.

Ablation study

In this section, this paper investigates a clustering-based scheme in a scenario in which a number of clients train the load flow prediction problem collaboratively. The accuracy of the proposed proposed model based on clustering is shown in Table 2 when the cluster center $K = 0, 2, 4, 8, 16$. Improve the original model's performance even more. When $K = 8, 16$, the cluster-based model can even surpass the centralized GRU model compared to the GRU model, which still jeopardizes data privacy. Because the clustering scheme

Table 2 Performance comparison on cluster settings

Methods	MAE	RMSE	MAPE
GRU	0.1123	0.1466	2.6022%
BFTL(Ours) ($K = 1$)	0.1158	0.1590	2.6148%
BFTL(Ours) ($K = 2$)	**0.1120**	0.1542	**2.6010%**
BFTL(Ours) ($K = 4$)	0.1061	**0.1431**	2.4701%
BFTL(Ours) ($K = 8$)	0.1033	0.1399	2.4133%
BFTL(Ours) ($K = 16$)	0.0997	0.1338	2.3536%

can group similar spatiotemporal features into a certain cluster, the best global model set can be integrated. As a result, the proposed proposed federated learning scheme based on clustering can enhance prediction performance even further, resulting in more accurate and responsive load flow prediction.

Discussion

Limitations

Geographic Bias: Training data is limited to Dutch charging patterns (EVnetNL dataset), which may not generalize to regions with differing EV adoption rates or grid infrastructures.

Computational Overhead: Homomorphic encryption introduces latency (Paillier: 148 ms/key vs. 22 ms for AES), making real-time forecasting challenging for sub-hourly granularity.

Threat Model Simplifications: Assumes honest-but-curious participants–malicious clients could theoretically disrupt consensus via Sybil attacks, though blockchain's PoA mitigates this risk.

Practical advantages in electric vehicle applications

The proposed framework delivers three critical enhancements for EV charging infrastructure that address unique sector challenges. First, it directly mitigates privacy risks inherent in location-based charging data by implementing homomorphic encryption during model aggregation (Sect. Cryptographic implementation)—a crucial advancement given that 72% of public charging sessions expose precise geolocation data vulnerable to pattern analysis. Second, the transfer learning component (Sect. 4.3) demonstrates 28.8% lower MAE than conventional approaches when handling stochastic charging behaviors (Table 1), effectively adapting to regional variance in urban charging patterns observed across COROP regions. Third, blockchain-secured model updates (Sect. 4.4) prevent manipulation of charging forecasts – a critical defense mechanism against the 37% increase in EV charger cyberattacks documented since 2023. By validating on the EVnetNL dataset containing 1.7 million real-world transactions, the framework proves particularly suited for large-scale deployments where traditional federated learning struggles with communication overhead (61.29% reduction at c=10 clients). These innovations collectively address the EV industry's dual challenge of maintaining grid reliability while protecting sensitive mobility data – an equilibrium unattainable through either differential privacy or centralized approaches alone.

Conclusion

In conclusion, the proposed study presents a novel blockchain-based federated transfer learning framework for privacy-aware probabilistic EV load forecasting, addressing critical challenges in the rapidly growing electric vehicle sector. By combining secure parameter aggregation via blockchain for decentralized training and transfer learning for model adaptability, the proposed approach offers a robust solution that prioritizes both privacy protection and forecasting accuracy. Experimental results demonstrate the framework's superior performance compared to traditional methods, especially in scenarios with limited data availability. The proposed model not only outperforms conventional approaches within privacy constraints but also shows promise for large-scale implementation, making it well-suited to meet the evolving demands of EV infrastructure. As we look to the future, the integration of Graph Convolutional Networks (GCN) to capture temporal and spatial relationships in the data presents an exciting opportunity to further enhance the model's predictive capabilities. What's more, we commit to simulating gradient inversion attacks in future work while attack resistance is an emergent property of the proposed architecture. Ultimately, this innovative framework represents a significant advancement in balancing accurate EV load forecasting with stringent data privacy requirements, paving the way for more secure and efficient management of EV charging systems.

Author Contributions Ruochen Jin: Methodology, Investigation. Hong Liu: Reviewing, Supervision.

Declaration:

Conflict of interest The authors declare no Conflict of interest. This work was supported in part by the National Natural Science Foundation of China under Grant 62072184.

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